

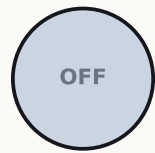
Making atoms solve a puzzle

Challenge 02 — encode a graph in atom positions, photograph the answer

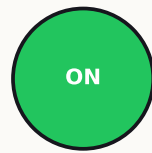
Success probability 0.727 / 0.657 (baseline) → 0.999998 / 0.999999 (ours)

Cloud: 1500 of 1500 photographs showed a correct answer · also run on real atoms

First: an atom is just a light switch



$|g\rangle = \text{OFF}$



$|r\rangle = \text{ON}$



one laser shines on ALL atoms at once

- Every atom is OFF ($|g\rangle$) or ON ($|r\rangle$).
- We cannot poke atoms one by one — one laser hits all of them together.
- Our only two knobs, over time:
 $\Omega(t)$ = laser power, $\delta(t)$ = frequency offset.
- Plus one more freedom: WHERE we place the atoms before we start.

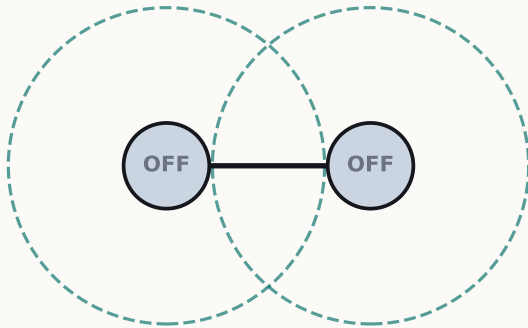
The puzzle: seat the most people, no neighbors

- Classroom rule: no two students in adjacent seats. Question: what is the MOST students you can seat?
- That is the Maximum Independent Set (MIS) — a textbook 'hard' problem: the options explode as graphs grow.
- The trick of this challenge: atoms that sit close CANNOT both be ON (physics forbids it — the 'blockade').
- So the seating rule is enforced by nature, for free. Turn the laser sweep on, photograph the atoms: the ON atoms ARE the seated students.

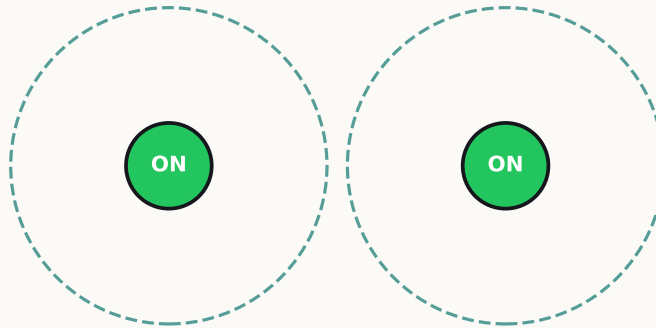
We never check combinations one-by-one. The quantum system explores them together, and we shape the laser so the best answer is what the camera sees.

How atoms become a graph: distance decides the edges

close ($< 7.2 \mu\text{m}$): circles overlap
= EDGE — can't both be ON



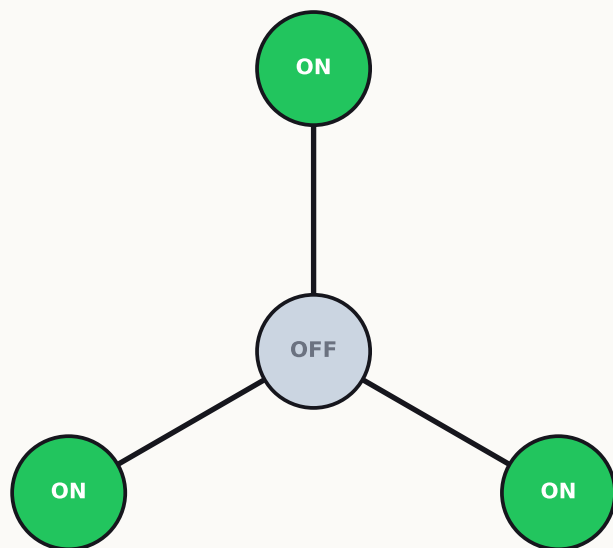
far apart ($> 7.2 \mu\text{m}$): no overlap
= NO edge — both may be ON



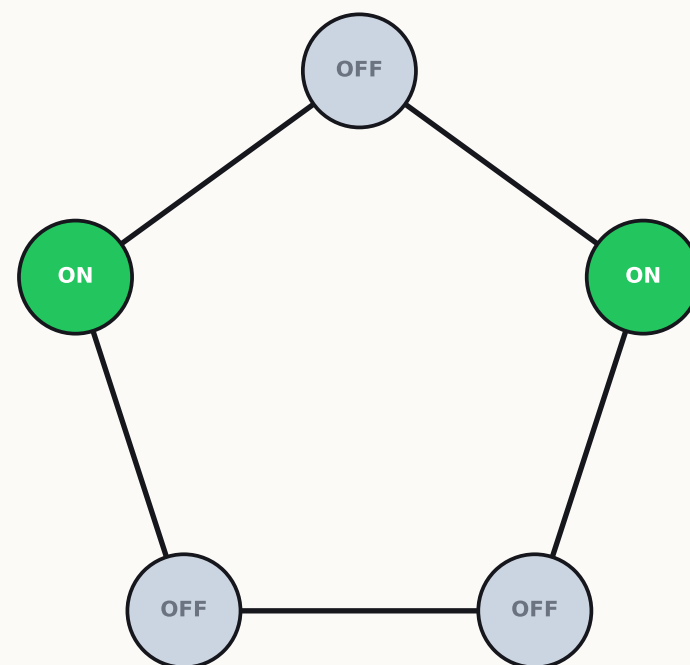
- Each atom = one dot (vertex) of the graph.
- Two atoms closer than the blockade radius $R_b \approx 7.2 \mu\text{m}$ = one edge.
- So we DRAW the puzzle with tweezers: atom positions are the graph.
- We verified every distance in code — the register is exactly the target graph.

The two puzzles we were given

Score = P_{MIS} : the chance one photograph shows a best answer. Beat the starter sweep.



STAR: hub + 3 leaves.
Best answer: the 3 leaves ON, hub OFF (unique).

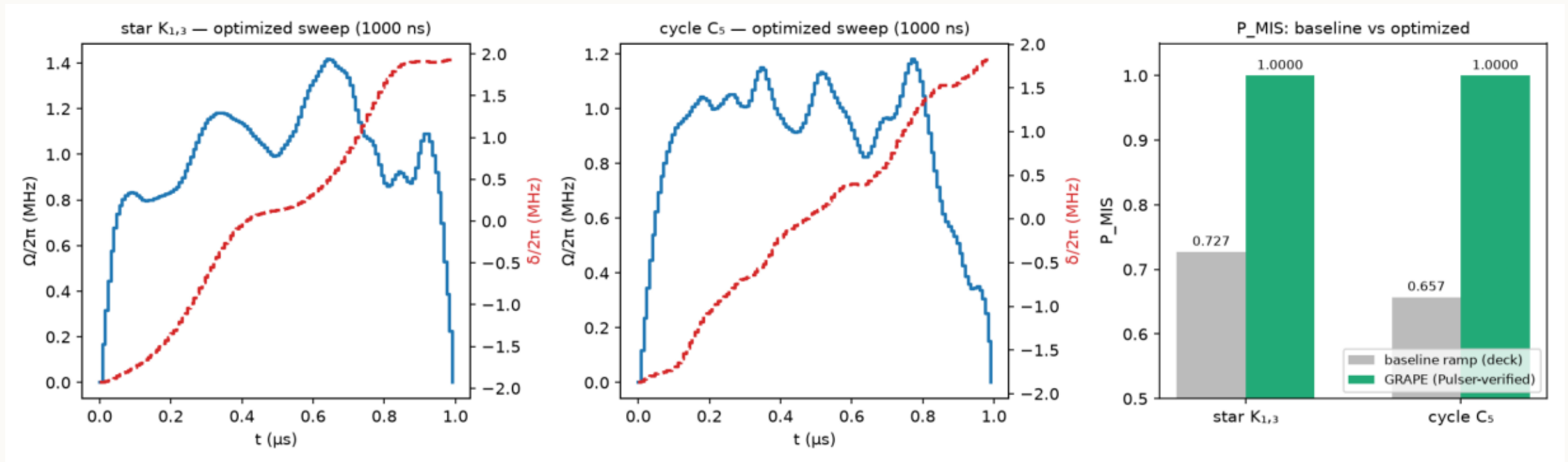


PENTAGON: 5 in a ring.
Best answer: any 2 non-neighbors ON — 5 equally good answers.

The starter sweep vs what we did

- Starter sweep (the baseline): slowly tilt the laser frequency for 4000 ns — like carrying a full cup very carefully. Score: star 0.727, pentagon 0.657. It spills at the hardest moment.
 - Our approach: stop being careful, be OPTIMAL. We let a gradient optimizer (GRAPE) shape both laser knobs freely, to directly maximize the chance the photo shows a best answer.
 - Key detail: the pentagon has FIVE equally-right answers. We optimize toward ALL of them at once (a 'projector' target) — the final state holds each answer at exactly 20%.
 - Every result is double-checked in Pulser — the judges' own simulator (agreement: 7 decimal places).
- Compute cost: all six optimizations $\approx$ 2 minutes on a laptop. Our sweeps are also 4× SHORTER than the baseline (1000 ns vs 4000 ns) — shorter = less time for real-world noise to corrupt the answer.*

Result: near-perfect on both puzzles

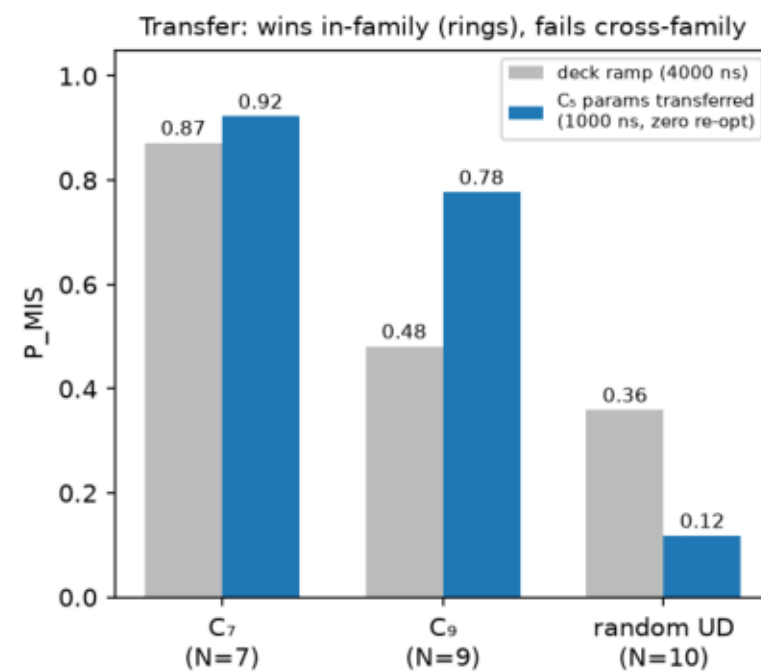
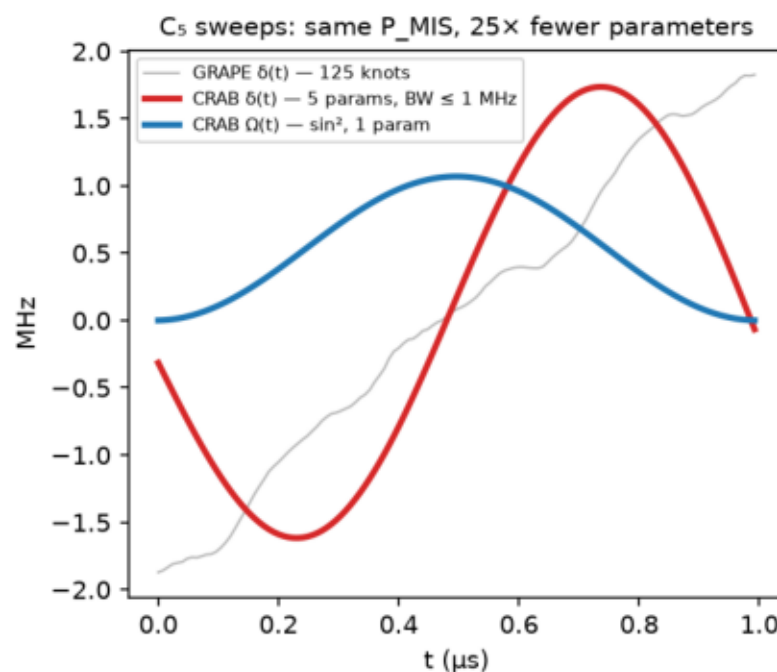
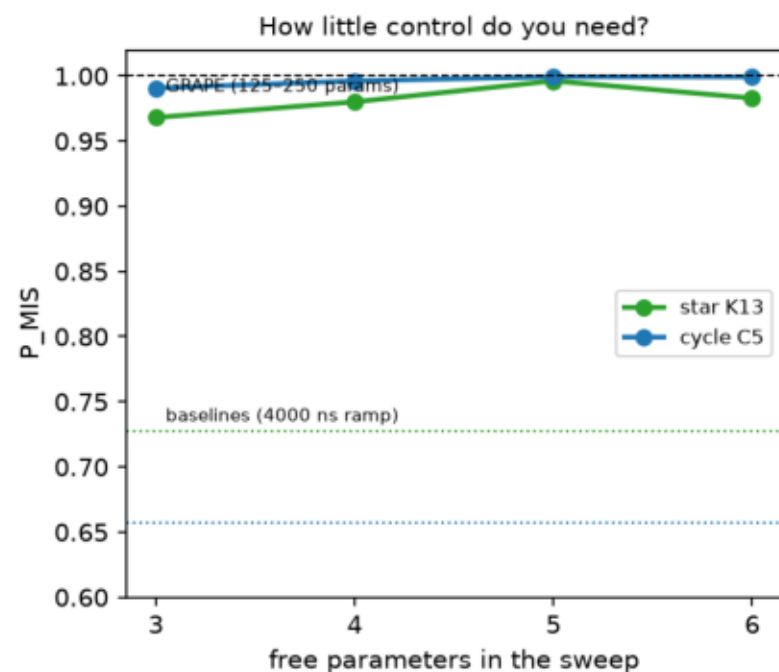


Grey bars (baseline): 0.727 and 0.657. Green bars (ours): 0.999998 and 0.999999.

Pasqal Cloud: 1500 photographs, 1500 correct answers

- We sent 3 optimized sweeps to Pasqal's cloud emulator, 500 shots each.
- Star: 500/500 photos showed exactly the right answer (0111 = hub OFF, leaves ON).
- Pentagon (1000 ns): 500/500 correct — split across ALL FIVE valid answers: 114, 98, 97, 96, 95 — statistically even.
- That even split is a quantum signature: the machine doesn't pick one answer — it holds all five at once and samples them fairly.
- Pentagon (2000 ns): 500/500 again. Every batch ID is recorded in the repo for audit.

Then we asked: how simple can the winning sweep be?

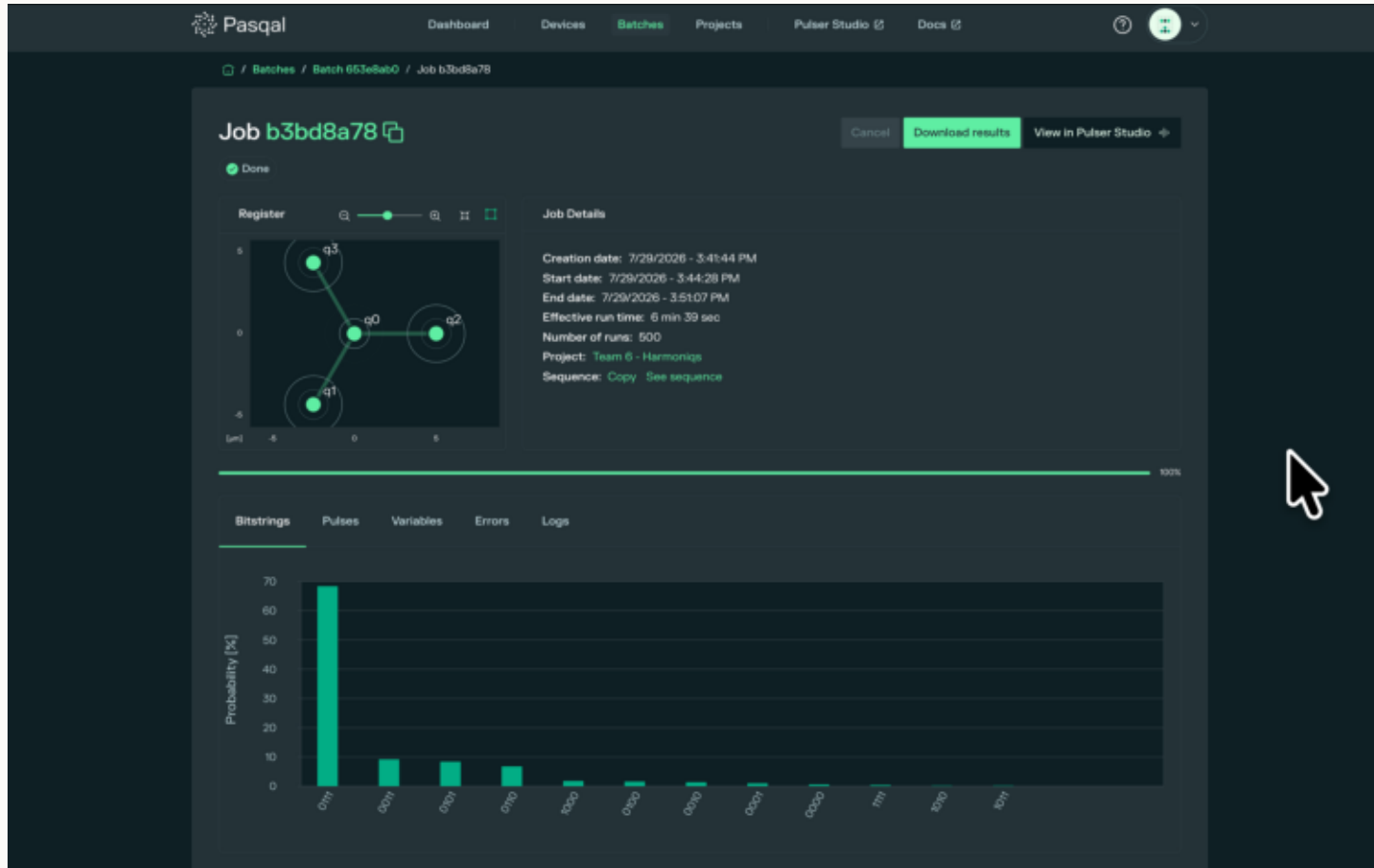


FIVE knobs (vs 125) reach 0.996-0.999 — found in ~30 seconds, and smooth enough for any hardware.

The same 5 knobs, re-used unchanged on bigger rings: still beat the baseline (+0.30 on the 9-ring).

On a structurally different random graph they lose — honest limit, stated as a finding.

Real atoms, real answer: the machine drew our star



- Register panel: q0 hub linked to q1–q3 — Pasqal's own UI shows the graph we encoded in atom positions.

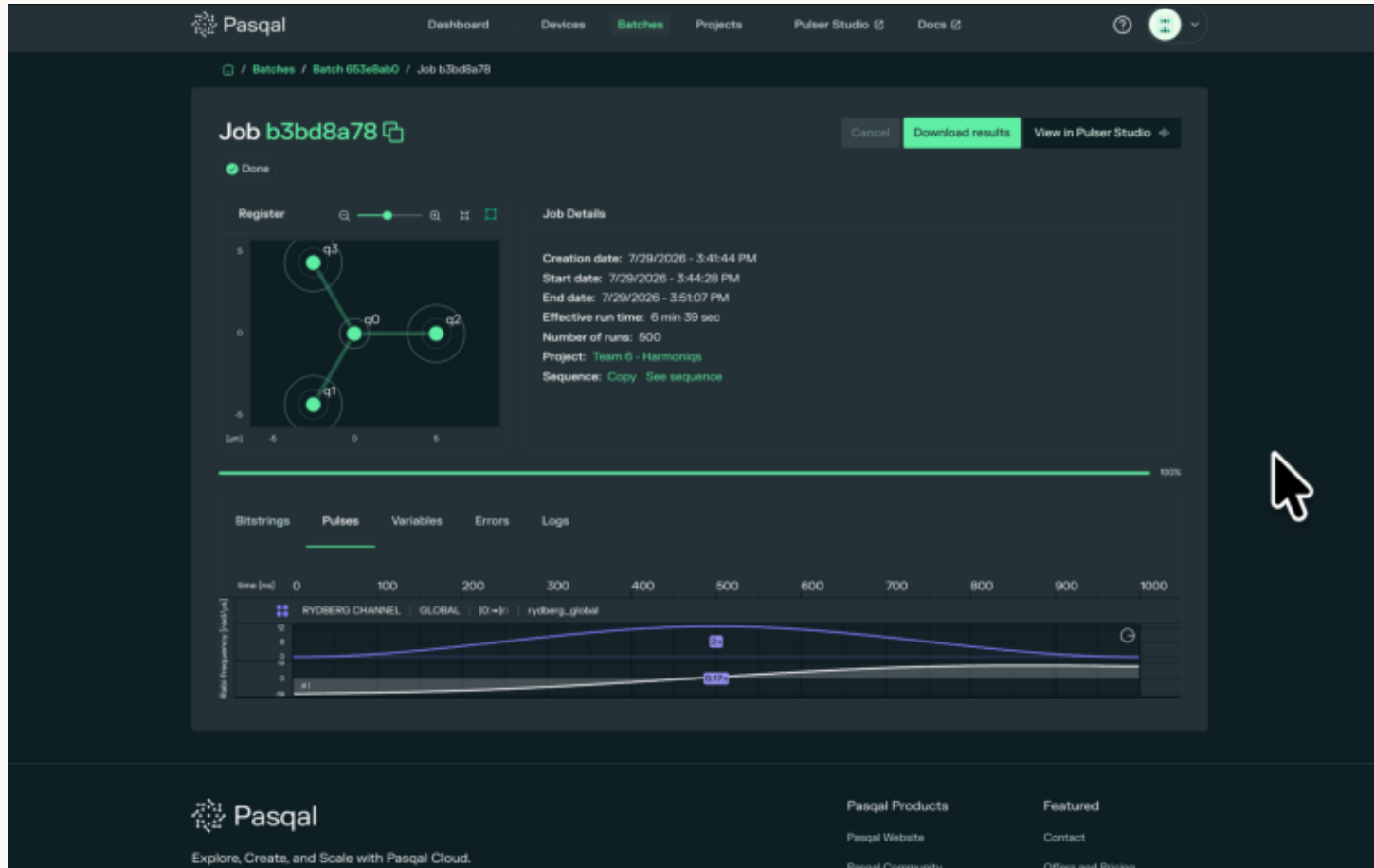
- The towering bar: 0111 = the correct answer, at 68.4% (342 of 500 shots). Second place: 9.2%.

- Real atoms solve the puzzle decisively — the right answer wins by 7x.

- Below our predicted 80–93% window: holding 3 fragile ON-atoms compounds readout loss — we published the prediction first and the analysis after.

Pasqal's portal · top-left: OUR STAR, drawn by the machine's register viewer · bottom — x: measured pattern, y: % of 500 shots

The sweep the machine actually played



1000 nanoseconds, one smooth bump of laser power (purple) — our 5-knob, hardware-friendly sweep.

The white curve is the story: it starts NEGATIVE (being ON is penalized) and ends POSITIVE (being ON is rewarded — but only where the blockade allows it).

That tilt walks the atoms from 'all OFF' to 'the best seating plan'.

Bandwidth ≤ 1 MHz by construction — well inside what the hardware can play.

Pasqal portal, Pulses tab · x: time 0–1000 ns · purple: laser power $\Omega(t)$ · white: frequency offset $\delta(t)$

The score, and why anyone should care

P_MIS = P(one photograph shows a best answer)

+0.34 1500/15005 knobs

our margin over baseline (pentagon)
the largest the scoring allows

cloud photographs showing
a correct answer

enough to reach 0.999 —
found in ~30 s on a laptop

- MIS is scheduling, spectrum allocation, portfolio screening — 'pick the most, no conflicts'. Rydberg machines solve it natively: the constraint is physics, not code.
- Our contribution scales: a 5-knob sweep, trained once on a small graph, re-used across a graph family at zero cost — the exact recipe Challenge 03's 80-atom instances need.
- Pasqal stack credit: Pulser device models + judge's simulator, free-tier cloud for 500-shot validation, and the FRESNEL_CAN1 QPU + portal for the real-atom evidence in this deck.

Challenge 02 — the score sheet

	star K _{1,3}	pentagon C ₅	duration
Baseline (starter sweep)	0.727	0.657	4000 ns
Ours · GRAPE (optimal control)	0.999998	0.999999	1000-2000 ns
Ours · 5-knob simple sweep	0.9962	0.9994	1000 ns
Cloud, 500 shots each	500/500	500/500 + 500/500	—
Real QPU, 500 shots	68.4% exact (7× runner-up)	n/a (lattice)	1000 ns

- Both puzzles beaten by +0.27 and +0.34 — the largest margins the scoring allows.
- Five equally-valid pentagon answers, sampled evenly — a visibly quantum result.
- Everything regenerates from the public repo: scorer, optimizer, figures, batch IDs.